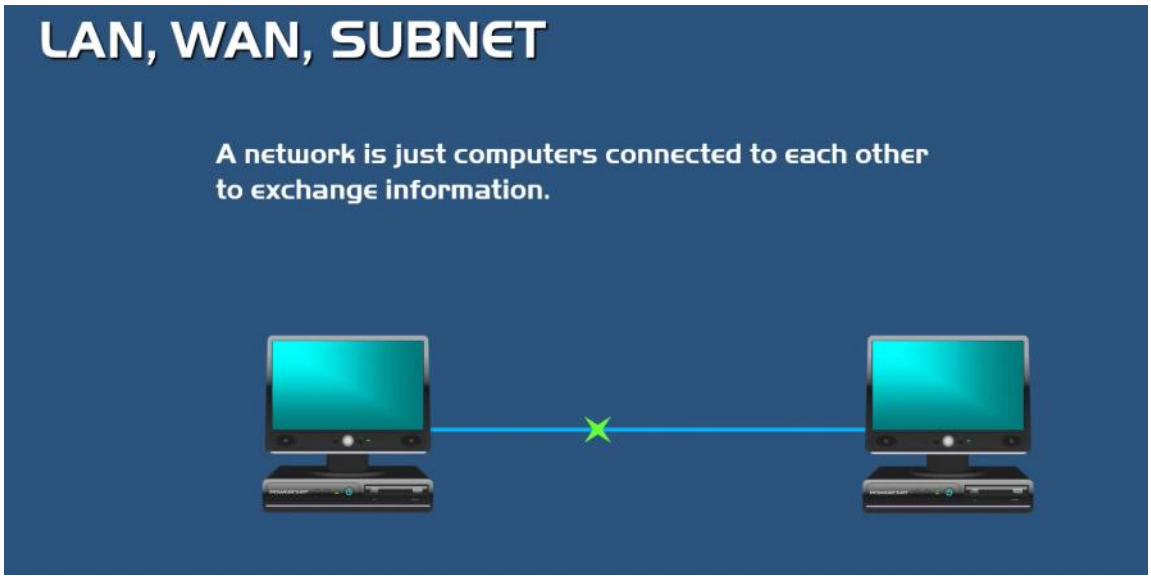
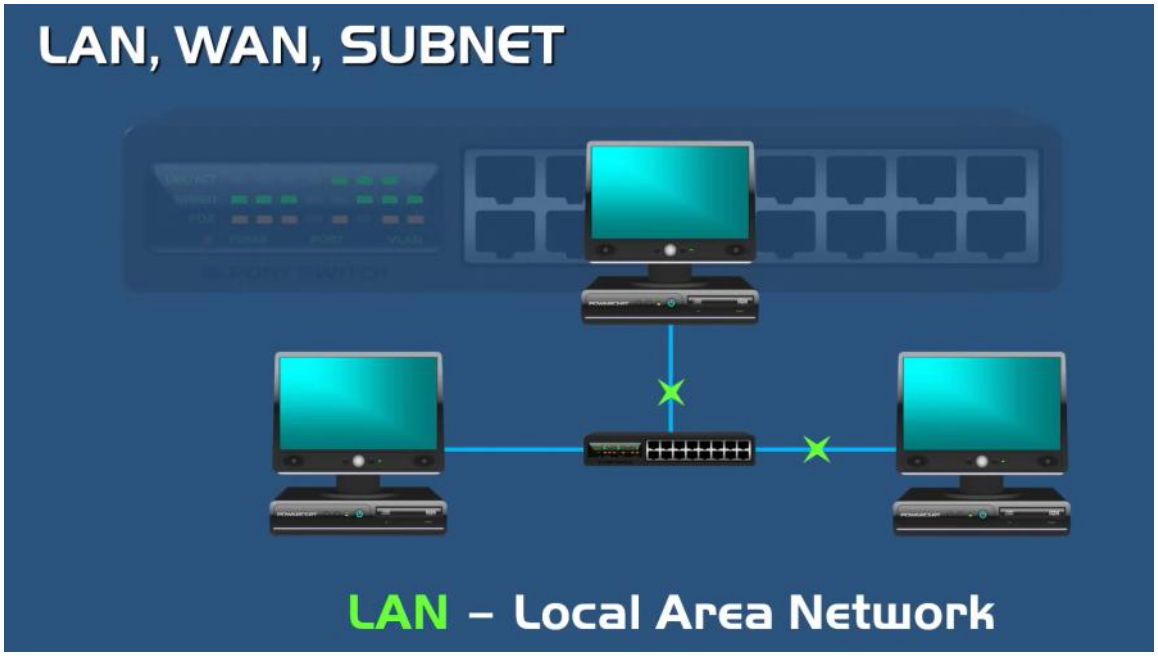
LAN, WAN, SUBNET

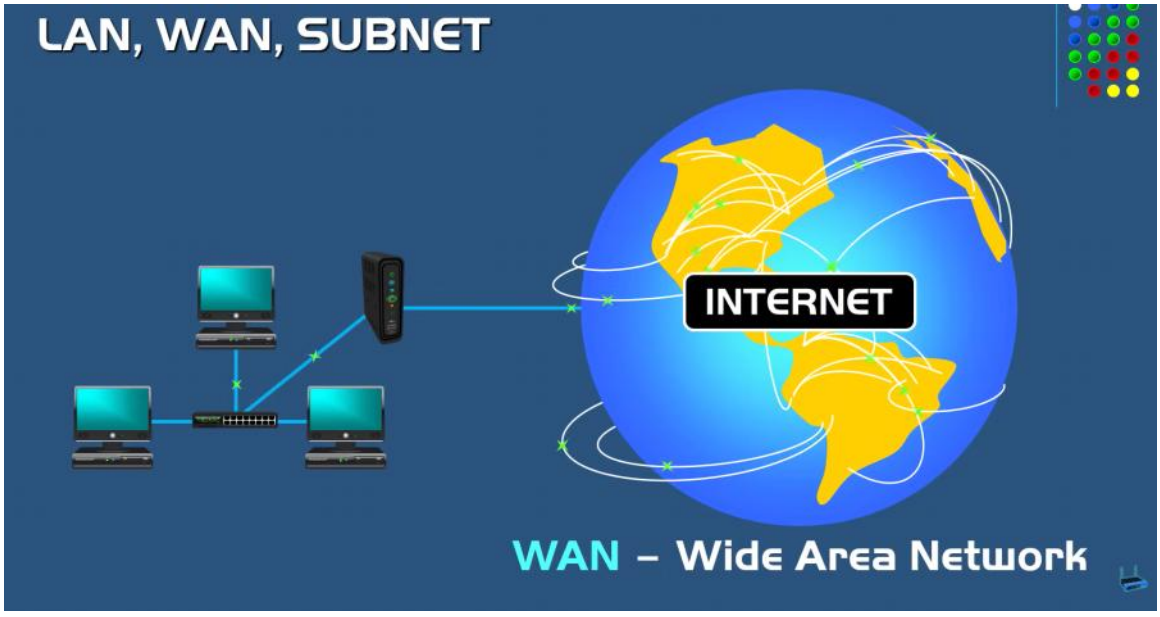
Thursday, April 9, 2026 12:54 PM

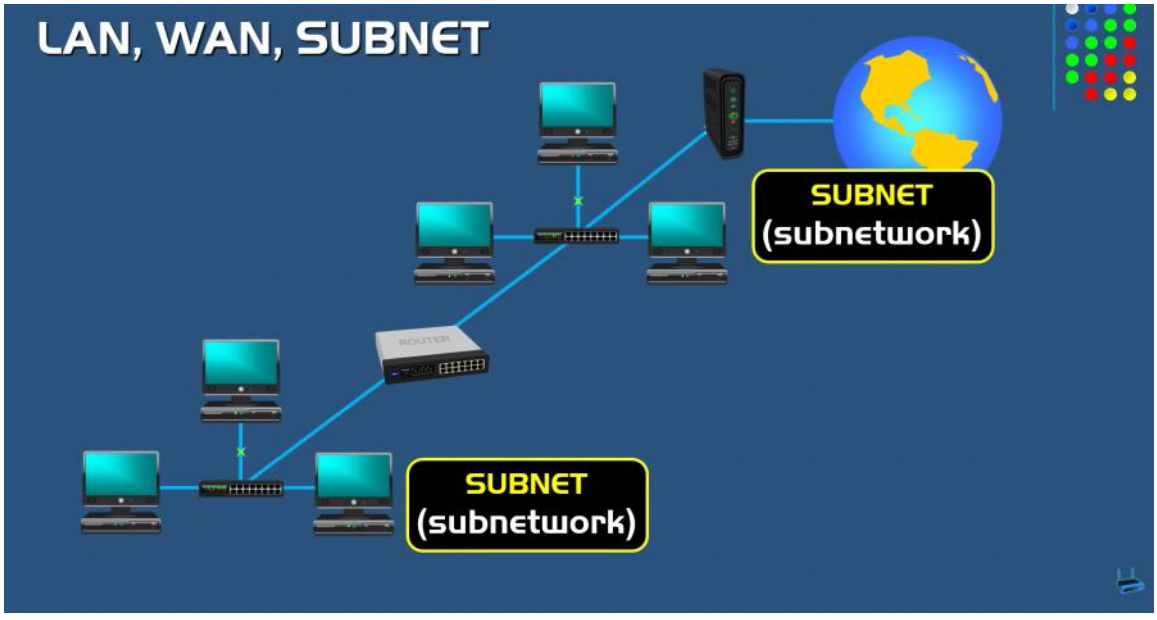
Simple Summary: LAN, WAN, Subnet

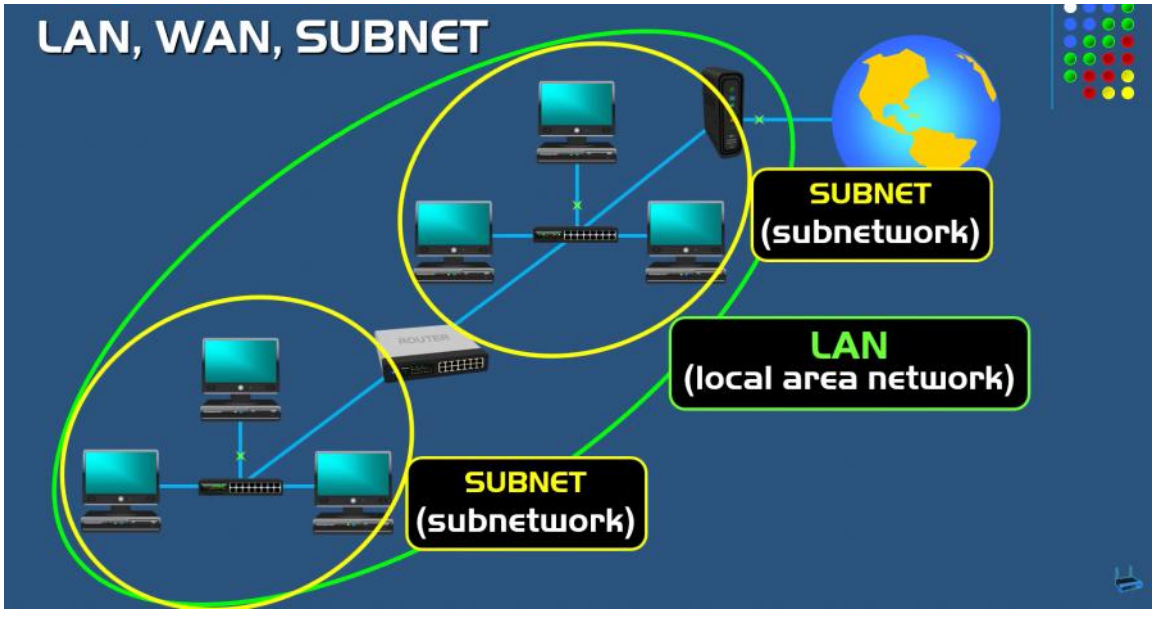
- A **Network** is a group of computers connected to share data.
- A **LAN (Local Area Network)** is a small private network inside a home, office, or building.
- A **WAN (Wide Area Network)** is a large network that connects multiple LANs over long distances (example: the internet).
- A **Subnet** is a smaller division inside a LAN used to organize and control network traffic.

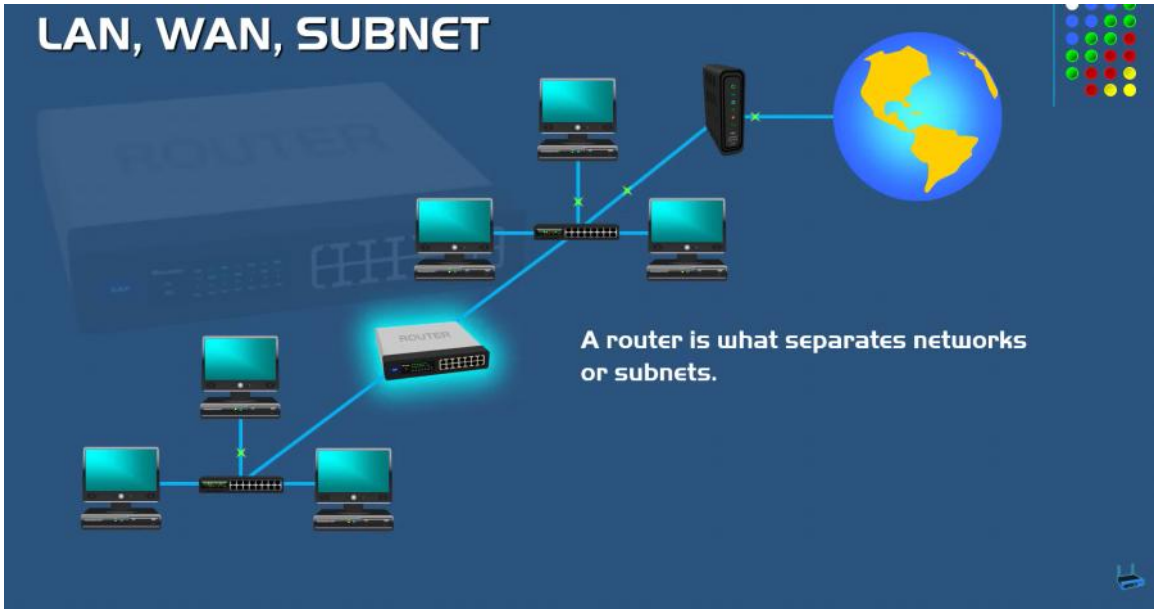




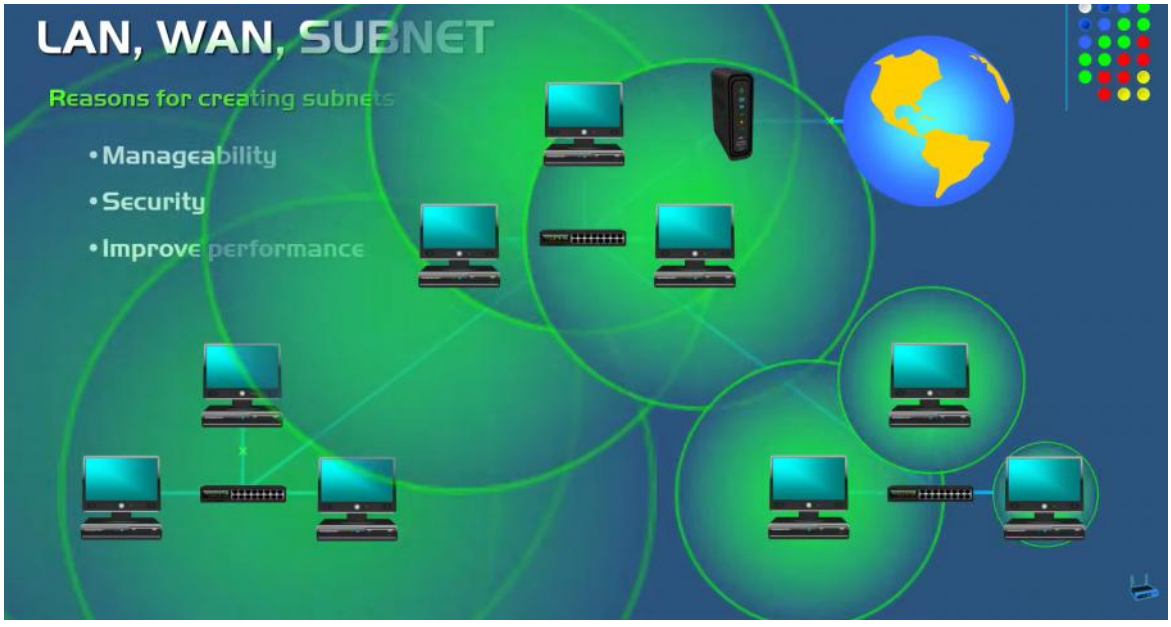

LAN, WAN, SUBNET
INTERNET
WAN – Wide Area Network


LAN, WAN, SUBNET
SUBNET
(subnetwork)
ROUTER
SUBNET
(subnetwork)

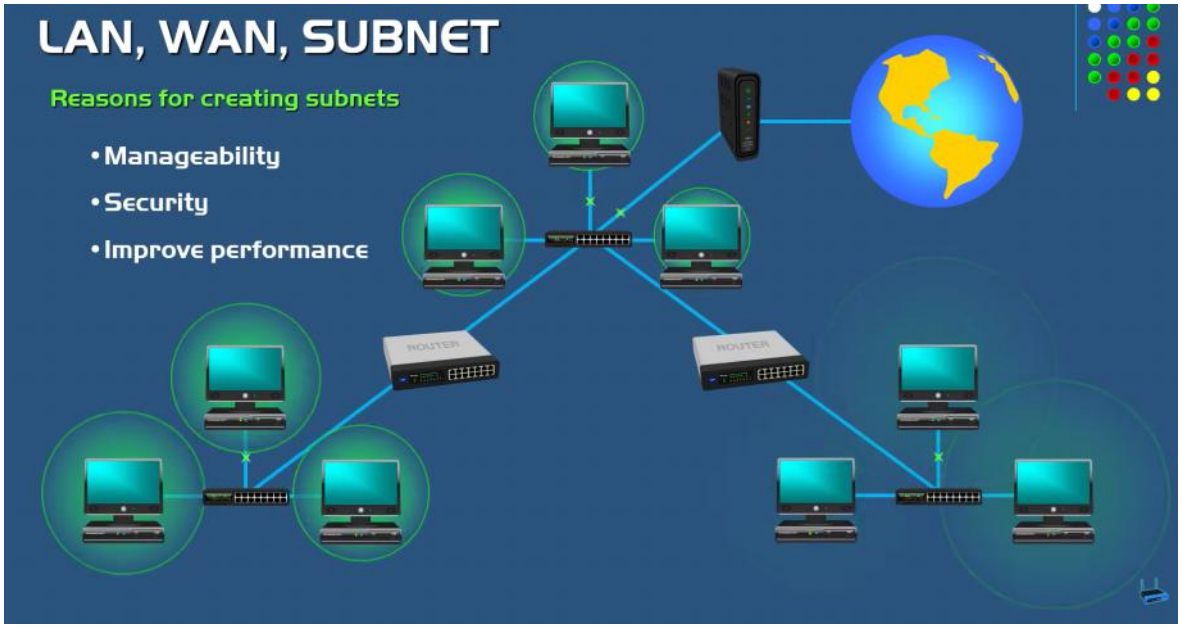

LAN, WAN, SUBNET
SUBNET
(subnetwork)
LAN
(local area network)
ROUTER
SUBNET
(subnetwork)



Before subnetting - The broadcast in sent to all the devices in NW



After subnetting - The broadcast is sent only to the devices in the same subnet



Why subnets are used

- Better management
- Improved security
- Reduced network traffic and better performance

Key idea:

WAN contains LANs, and LANs can be divided into subnets.